

HW5: NoSQL R&D Project (GROUP PROJECT)

Due Monday by 11:59pm

Points 100

Submitting an external tool

NoSQL R&D Project Specification

Report due: MONDAY June 15th @ 11:59PM

You may work individually or in groups of up to 5 people.

MOTIVATION

It can be difficult to find a good book on general NoSQL principles. The technologies continue to evolve rapidly and many of the popular books, such as "Seven databases in Seven Weeks" are very out-of-date. It would be great if there was a book called: **NoSQL 2026: Databases and Data Science Applications**. For HW5, the last assignment of the course, you will work collectively to help write this book. (I'm the editor!). I'm hoping our book will be a useful reference for you all in the years to come.

PROJECT REQUIREMENTS

For your final mini-project R&D assignment you can do one of the following:

a) **Database Evaluation:** Pick any NoSQL database other than Redis, Mongo, Neo4J and write a 5+ page summary describing key principles, use-cases, and a

basic tutorial that conveys the essential ideas behind the database. Imagine you are evaluation technology for your company's development team. Would this database be useful? Is it easy to use? Is it scalable? What are the licensing terms? Is there an active community of users? **GO BEYOND THE PRODUCT MARKETING!** Be objective and informative. Include a tutorial demonstration to convey your database's essential features and capabilities. Your evaluation (including tutorial) should be about 5+ pages.

b) Data Science Application: We've seen in class how NoSQL can be used as a back-end for a social network or to carry out bioinformatics research. Use a NoSQL database to implement an application or carry out a data science investigation. Document your data sources and methods. Include insightful visualizations and clearly state your conclusions. The focus of your investigation should have some scientific, cultural, or economic significance. Aim for innovation and originality. Write a 5+ page report. Submit your report and code.

Whether you choose (a) or (b), please prepare a PowerPoint presentation an in-class NoSQL conference to be held Tuesday June 16th and Wednesday June 17th. (The book will essentially be our class conference proceedings.)

PROJECT DECLARATION

Declare your project intentions [HERE](#)

[Links to an external site.](#)

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REPORT FORMAT AND GUIDELINES

A report written by AI will not be well received. ***Do not use AI to write prose! Don't let AI rob you of your voice.*** Single Your final report should be 5 or more pages, single spaced, including all tutorials, diagrams, and visualizations. The outline of your report depends on your topic:

Database Investigations

Title

Full names of all team members in alphabetical order by last name

Abstract

Introduction

Use-Cases

Tutorial demonstrating the use of the database and its query language

Strengths and Weaknesses

Conclusions or Summary

Research or Application Report:

Title

Full names of all team members in alphabetical order by last name

Abstract

Introduction

Methods

Analysis

Conclusions or Summary

WHAT TO SUBMIT TO GRADESCOPE:

- **5+ page report / evaluation / analysis (PDF)**
- **Data (zipped) if practical (ZIP)**
- **5 slide powerpoint presentation you will deliver in class on June 16th or June 17th (PDF)**
- **Code (if applicable)**

One Gradescope submission per project please. Submitter: Be sure authorship of all files is clear and that you have added all team members to the GradeScope submission. You will also be asked to declare how you used AI in completing the assignment and to document *in detail* how each group member contributed to the final result.

**ALL TEAM MEMBERS SHOULD VERIFY THAT THE PROJECT IS
CORRECTLY AND COMPLETELY SUBMITTED BY THE DUE DATE. THERE
ARE NO LATE EXTENSIONS! PROJECT REGRADE REQUESTS WILL NOT
BE CONSIDERED.**